DocMind System Architecture & Neural RAG Specifications
Technical Whitepaper & System Design Document
Version 2.0 - October 2026

1. EXECUTIVE OVERVIEW
DocMind is a high-throughput, local-first document intelligence platform engineered for zero-hallucination factual question a

2. SYSTEM ARCHITECTURE & HIGH-LEVEL PIPELINE
The DocMind core pipeline consists of four discrete processing stages:
- Ingestion & Text Extraction: PyMuPDF high-speed binary parsing preserving layout, page demarcations, and bounding bo
- Recursive Chunking Engine: Sliding-window tokenization with configurable chunk sizes (700 characters) and sliding overla
- Dense Neural Embedding: SentenceTransformers (all-MiniLM-L6-v2) mapping text passages into a 384-dimensional dens
- Vector Similarity & Retrieval: In-memory cosine similarity and FAISS index topology computing dot products between norm

3. PERFORMANCE BENCHMARKS & LATENCY TARGETS
DocMind delivers sub-50ms vector similarity lookup across 10,000 document chunks and sub-200ms end-to-end response

4. GROUNDED CITATION FORMULATION & ZERO-HALLUCINATION PROMISE
Unlike standard probabilistic language models that hallucinate plausible but fictional citations, DocMind enforces rigorous ev
- Exact Passage Injection: Retrieved chunks exceeding minimum cosine similarity thresholds (min_sim >= 0.25) are formatt
- Citation Provenance: The generative assistant is mathematically bound to synthesize answers solely from injected context
- Fallback Retrieval Guarantee: If no external LLM API credentials are configured, DocMind seamlessly transitions into a de

5. SECURITY, COMPLIANCE & LOCAL EXECUTION
All document ingestion, embedding computation, and vector indexing operations execute 100% locally within the private run

6. API CONTRACTS & INTEGRATION INTERFACES
DocMind exposes robust asynchronous REST and Server-Sent Event (SSE) endpoints:
- POST /api/v1/documents/upload: Multi-part binary ingestion with automatic indexing.
- POST /api/v1/chat/query: Grounded Q&A with top-k retrieved citations and response latency metrics.
- POST /api/v1/chat/search: Direct semantic clause and fact search returning similarity scores and relevance tracks.